

DESIGN AND IMPLEMENTATION OF A MULTI-AREA ENTERPRISE NETWORK USING DYNAMIC ROUTING, VPN AND REDUNDANCY TECHNOLOGIES

Scientific Coordinator:
Lect. Phd. Eng. Andy Vesa

Candidate:
Eng. Alex-Grațian Czinczas

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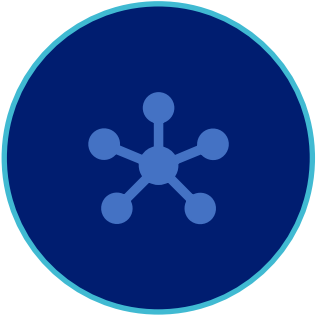
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1.PROJECT OBJECTIVES



Design a scalable enterprise network



Implement dynamic routing protocols

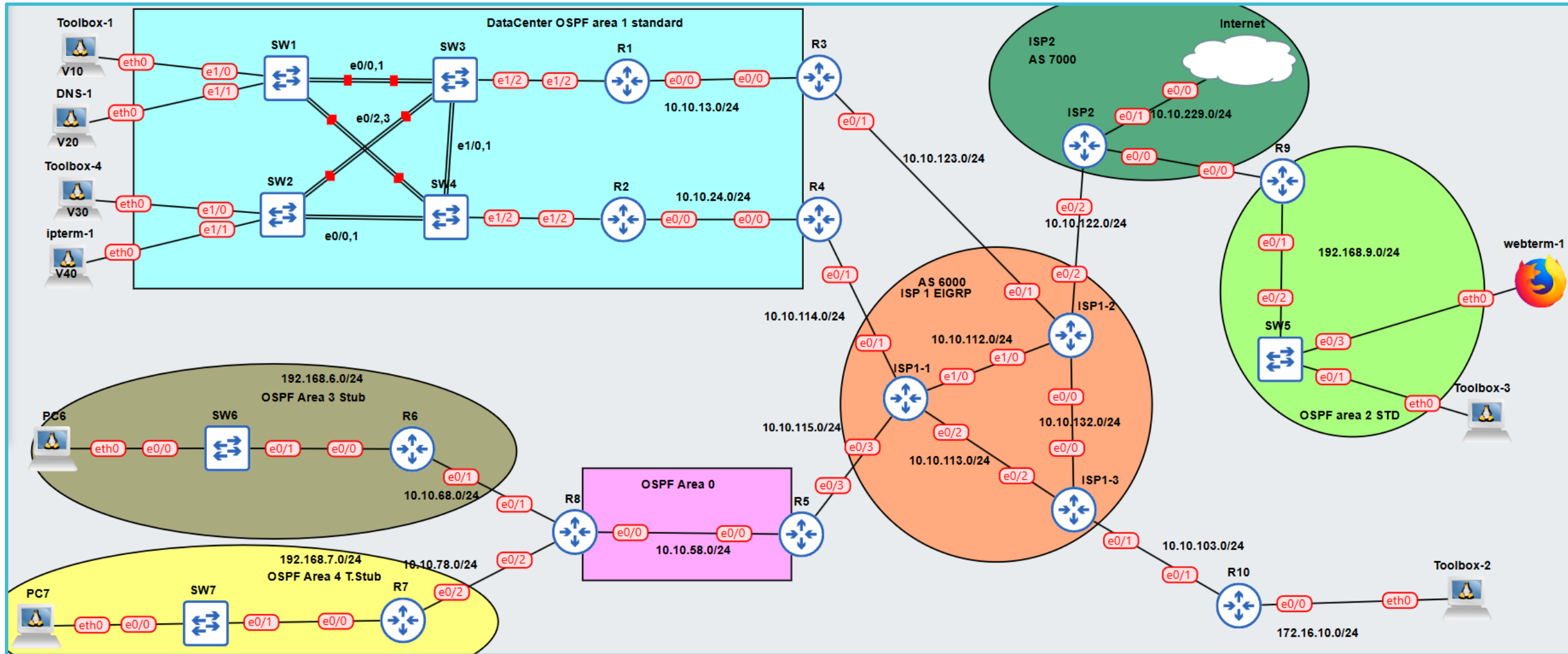


Ensure secure communication

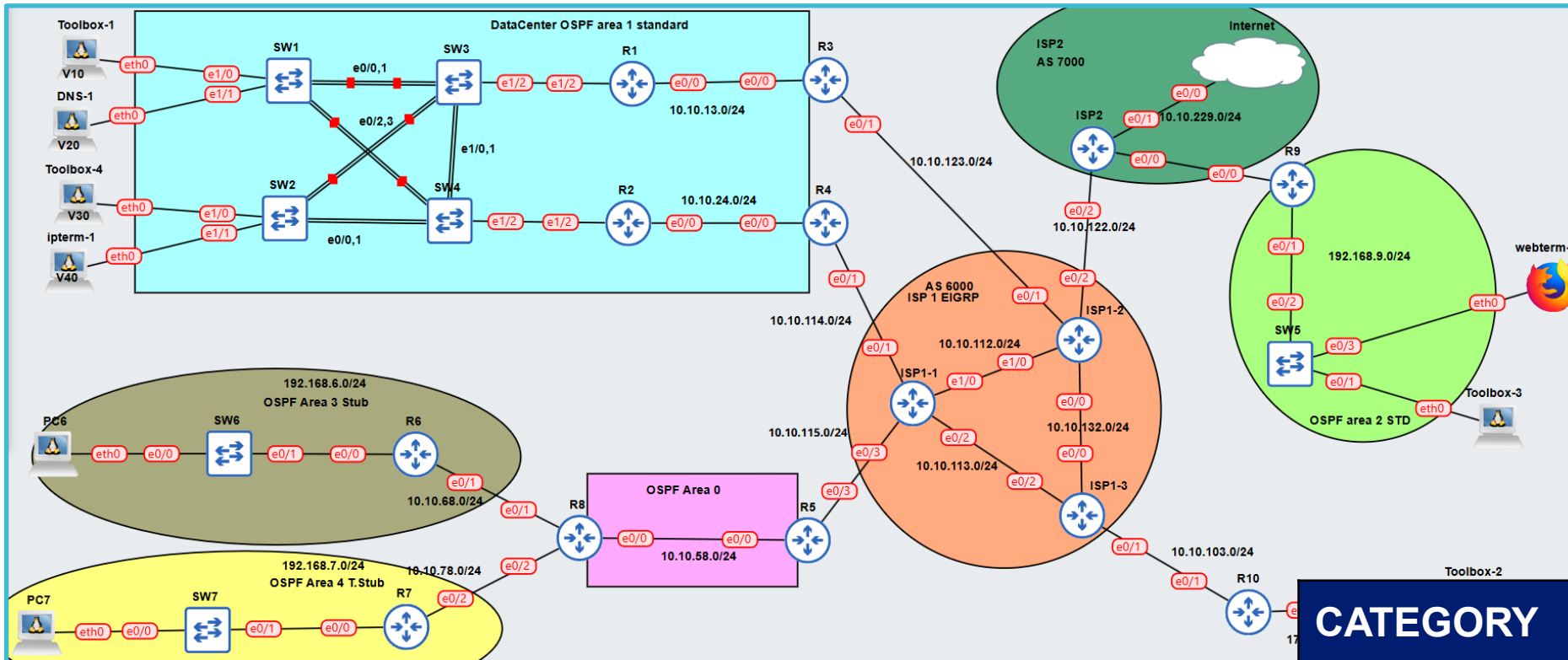


Provide redundancy and high availability

2.NETWORK TOPOLOGY



3.NETWORK DESIGN



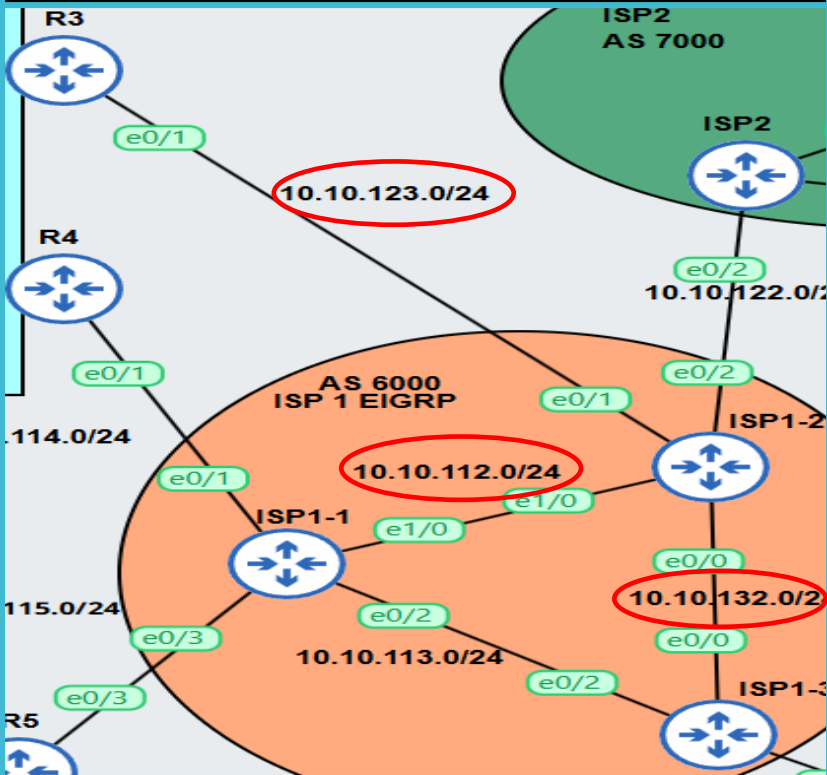
CATEGORY	TECHNOLOGIES
Routing	OSPF, EIGRP, BGP
VPN	GRE, IPsec
Redundancy	HSRP, EtherChannel, RPVST+
Network Services	DHCP+Relay, NAT/PAT

4.ROUTING IMPLEMENTATION

GNS3 console

ISP1-2 X

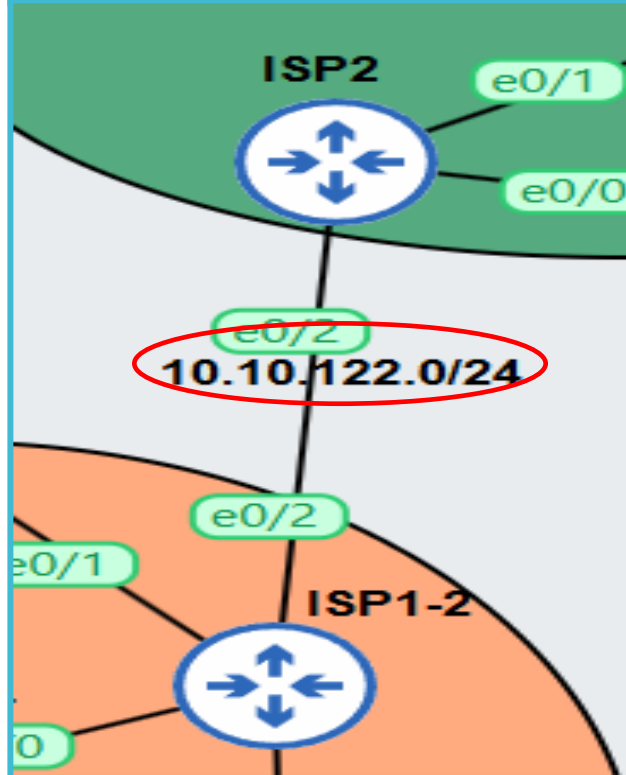
```
ISP1-2#show run | s eigrp
router eigrp 1
 network 10.10.112.0 0.0.0.255
 network 10.10.123.0 0.0.0.255
 network 10.10.132.0 0.0.0.255
 redistribute bgp 6000 metric 10000 1 255 1 1500
 redistribute eigrp 1
```



GNS3 console

ISP2 X

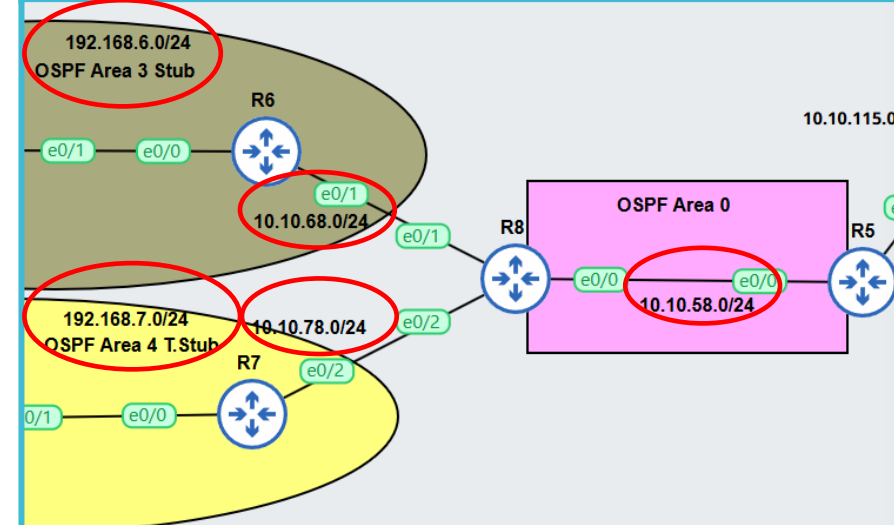
```
ISP2#show run | s bgp
router bgp 7000
 bgp log-neighbor-changes
 neighbor 10.10.122.2 remote-as 6000
 neighbor 10.10.122.2 description ISP1-2
 !
 address-family ipv4
  network 0.0.0.0
  redistribute connected
  neighbor 10.10.122.2 activate
 exit-address-family
```



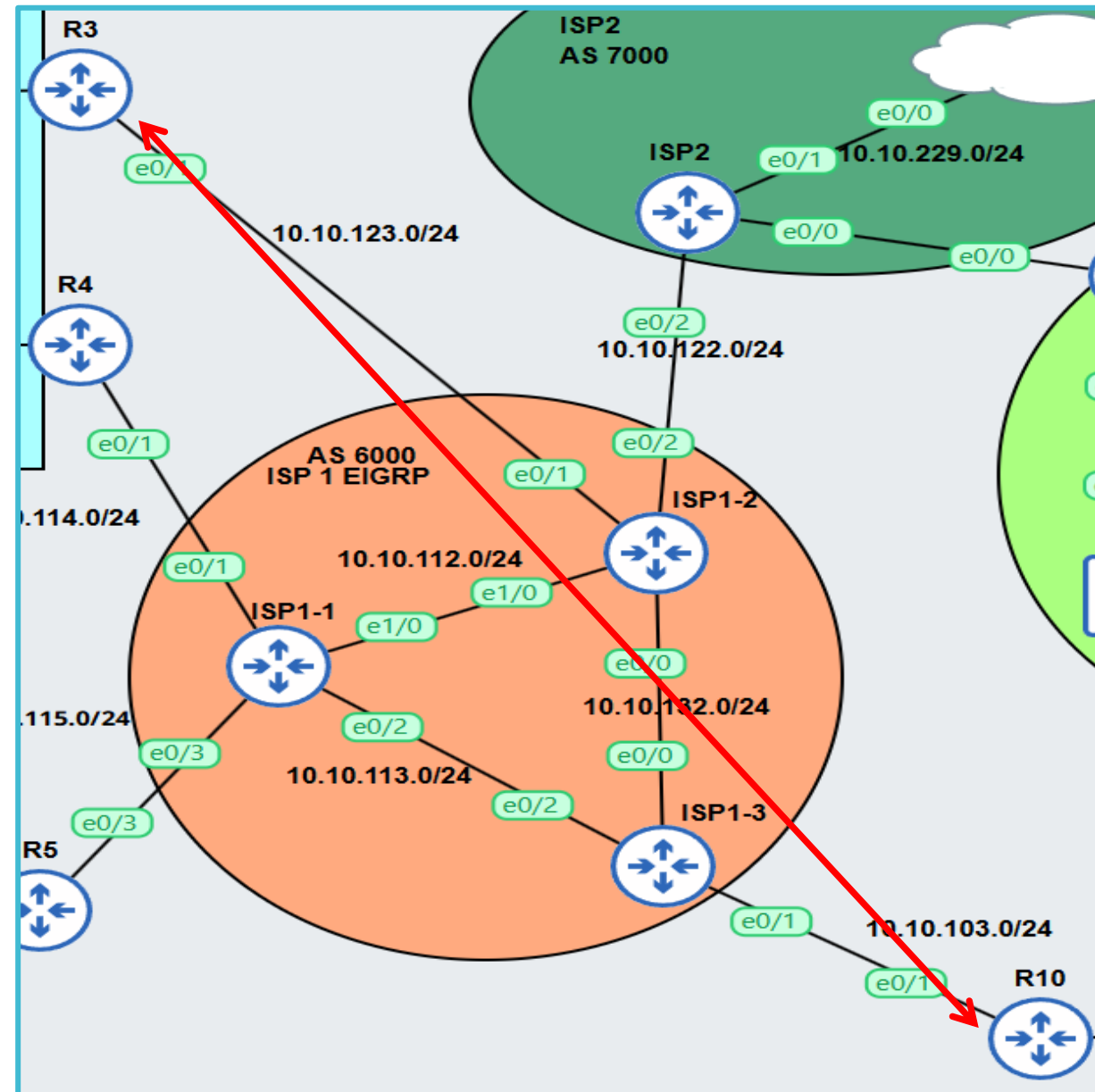
GNS3 console

R8 X

```
router ospf 1
 router-id 8.8.8.8
 area 3 stub
 area 4 stub no-summary
 network 10.10.58.0 0.0.0.255 area 0
 network 10.10.68.0 0.0.0.255 area 3
 network 10.10.78.0 0.0.0.255 area 4
```



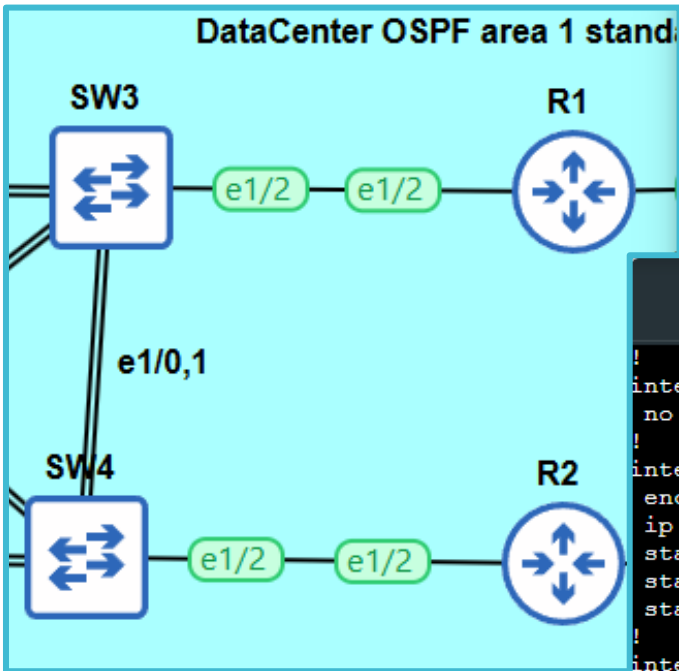
5. GRE OVER IPSEC



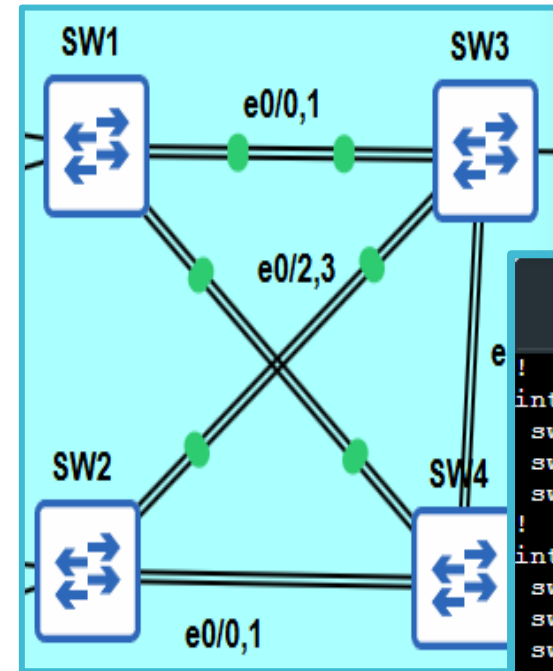
```
GNS3 console R3 X
R3#show run | b interface
interface Tunnel1
 ip address 10.255.60.1 255.255.255.252
 ip mtu 1400
 ip tcp adjust-mss 1360
 tunnel source Ethernet0/1
 tunnel destination 10.10.103.10

GNS3 console R3 X R10 X
R3#show run | b crypto
crypto isakmp policy 10
 encr aes
 authentication pre-share
 group 2
crypto isakmp key cisco address 10.10.103.10
!
!
crypto ipsec transform-set TS esp-aes esp-sha-hmac
 mode transport
!
!
!
crypto map CMAP 10 ipsec-isakmp
 set peer 10.10.103.10
 set transform-set TS
 match address GRE-R3-R10
```

6.HIGH AVAILABILITY



```
GNS3 console R1 X
!
interface Ethernet1/2
no ip address
!
interface Ethernet1/2.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
standby 10 ip 192.168.10.67
standby 10 priority 110
standby 10 preempt
!
interface Ethernet1/2.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
standby 20 ip 192.168.20.67
standby 20 priority 110
standby 20 preempt
!
interface Ethernet1/2.30
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
standby 30 ip 192.168.30.67
standby 30 priority 90
!
interface Ethernet1/2.40
encapsulation dot1Q 40
ip address 192.168.40.1 255.255.255.0
standby 40 ip 192.168.40.67
standby 40 priority 90
!
```



```
GNS3 console SW3 X
!
interface Port-channel3
switchport trunk encapsulation dot1q
switchport trunk allowed vlan 10,20,30,40
switchport mode trunk
!
interface Ethernet0/0
switchport trunk encapsulation dot1q
switchport trunk allowed vlan 10,20,30,40
switchport mode trunk
channel-group 1 mode active
!

GNS3 console SW1 X SW3 X
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
spanning-tree vlan 10,20 priority 24576
spanning-tree vlan 30,40 priority 28672
!
```

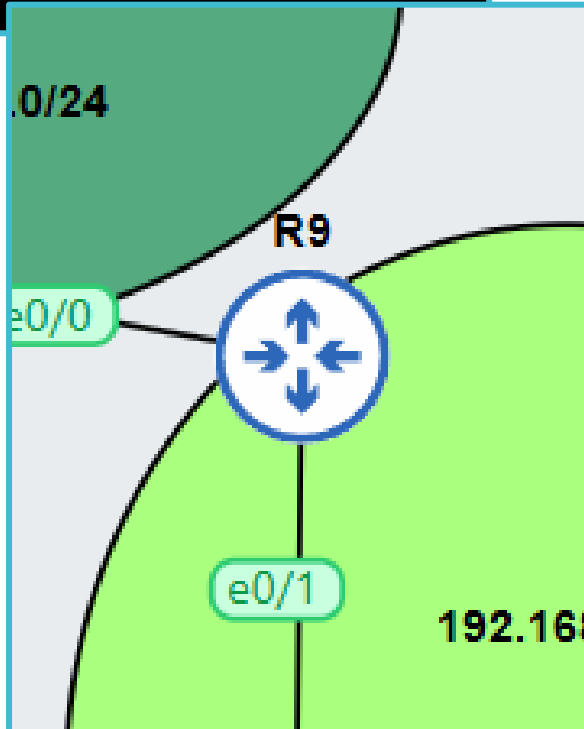

7.DHCP

GNS3 console

R6 X

R9 X

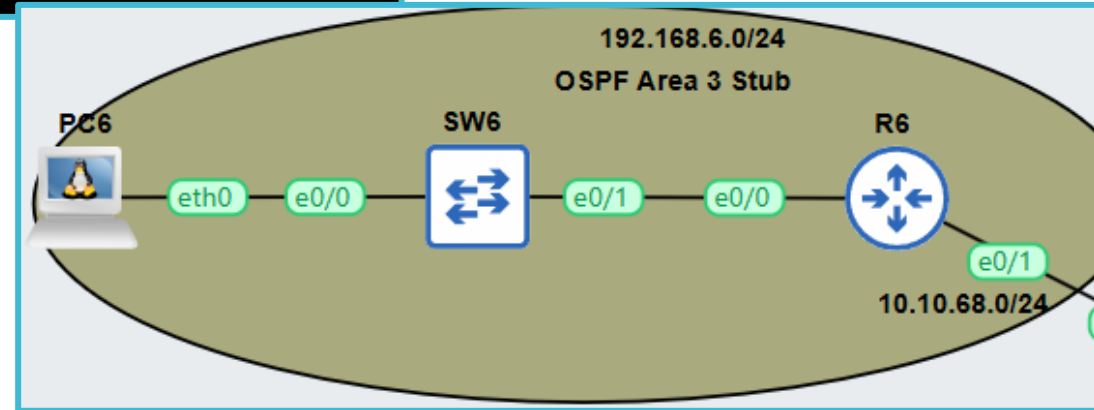
```
R9#show run | s dhcp
ip dhcp excluded-address 192.168.6.1 192.168.6.20
ip dhcp excluded-address 192.168.7.1 192.168.7.20
ip dhcp pool LAN_AREA3
 network 192.168.6.0 255.255.255.0
 default-router 192.168.6.6
 dns-server 8.8.8.8
ip dhcp pool LAN_AREA4
 network 192.168.7.0 255.255.255.0
 default-router 192.168.7.7
 dns-server 8.8.8.8
```



GNS3 console

R6 X

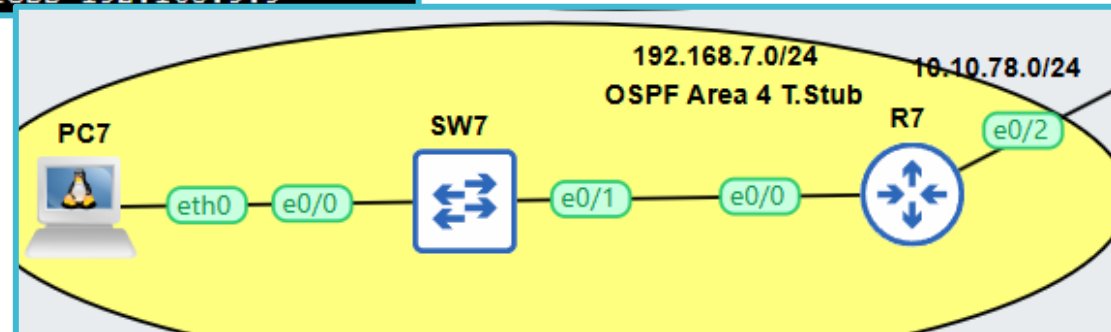
```
interface Ethernet0/0
ip address 192.168.6.6 255.255.255.0
ip helper-address 192.168.9.9
```



GNS3 console

R7 X

```
R7#show run | s 0/0
interface Ethernet0/0
ip address 192.168.7.7 255.255.255.0
ip helper-address 192.168.9.9
```



8.NAT/PAT

GNS3 console

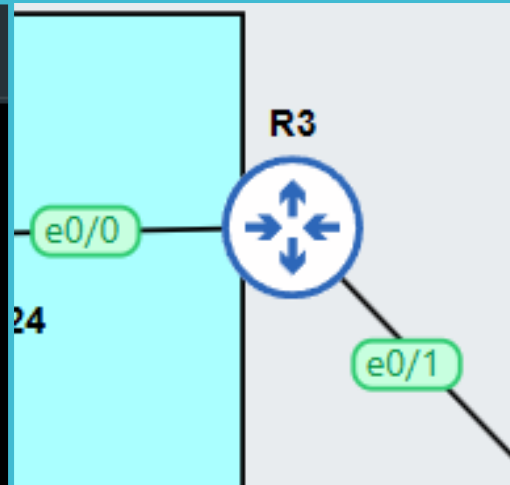
R3 ✕

```
ip nat inside source list NAT-R3 interface Ethernet0/1 overload
ip route 0.0.0.0 0.0.0.0 10.10.123.2
!
ip access-list standard NAT-R3
 permit 192.168.0.0 0.0.255.255
!
ip access-list extended GRE-R3-R10
 permit gre host 10.10.123.3 host 10.10.103.10
```

GNS3 console

R3 ✕

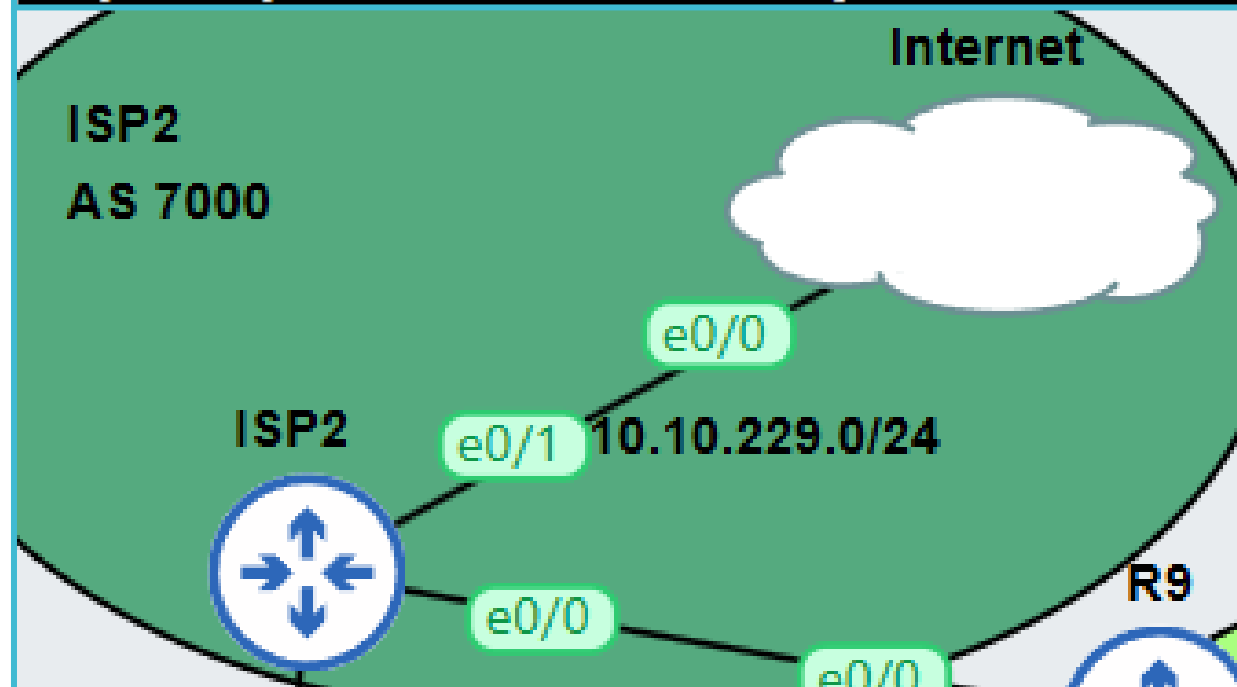
```
interface Ethernet0/0
 ip address 10.10.13.3 255.255.255.0
 ip nat inside
 ip virtual-reassembly in
!
interface Ethernet0/1
 ip address 10.10.123.3 255.255.255.0
 ip nat outside
 ip virtual-reassembly in
 crypto map CMAP
```



GNS3 console

ISP2 ✕

```
ip nat inside source list 100 interface Ethernet0/1 overload
ip ssh bulk-mode 131072
!
ip access-list extended 100
 10 permit ip 10.0.0.0 0.255.255.255 any
 20 permit ip 172.0.0.0 0.255.255.255 any
 30 permit ip 192.0.0.0 0.255.255.255 any
```



9.VALIDATION RESULTS

OSPF Adjacency

```
R8#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
5.5.5.5	1	FULL/BDR	00:00:36	10.10.58.5	Ethernet0/0
6.6.6.6	1	FULL/BDR	00:00:31	10.10.68.6	Ethernet0/1
7.7.7.7	1	FULL/BDR	00:00:32	10.10.78.7	Ethernet0/2

```
R8#
```

IPsec VPN

```
R5#show crypto isakmp sa
```

IPv4 Crypto ISAKMP SA		state	conn-id	status
dst	src			
10.10.115.5	10.10.229.9	QM_IDLE	1001	ACTIVE

```
IPv6 Crypto ISAKMP SA
```

EIGRP Adjacency

```
ISP1-2#show ip eigrp neighbors
```

EIGRP-IPv4 Neighbors for AS(1)						
H Seq	Address	Interface	Hold (sec)	Uptime	SRTT (ms)	RTO Q Cnt
0	10.10.132.3	Et0/0	11	00:00:36	1	100 0
1	10.10.112.1	Et1/0	13	00:00:36	5	100 0

```
ISP1-2#
```

GRE Tunnel

```
R5#show ip int br | s up
```

Interface	Address	Encapsulation	Hardware	Operational
Ethernet0/0	10.10.58.5	YES NVRAM	up	up
Ethernet0/3	10.10.115.5	YES NVRAM	up	up
Tunnel0	10.255.59.1	YES NVRAM	up	up

eBGP Adjacency

```
Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down St
```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	St
10.10.122.1	4	7000	83	92	93	0	0	01:12:15	

```
ISP1-2#
```

HSRP Redundancy

```
R1#show standby brief
```

Interface	Grp	Pri	P State	Active	Standby	Virtual IP
Et1/2.10	10	110	P Active	local	192.168.10.2	192.168.10.67
Et1/2.20	20	110	P Active	local	192.168.20.2	192.168.20.67
Et1/2.30	30	90	Standby	192.168.30.2	local	192.168.30.67
Et1/2.40	40	90	Standby	192.168.40.2	local	192.168.40.67

```
R1#
```

NAT Translation

```
R3#show ip nat translations | i 192.168.10.10
```

Protocol	Inside Local	Inside Global	Outside Local	Outside Global
icmp	10.10.123.3:2111	192.168.10.10:2111	10.10.122.1:2111	10.10.122.1:2111

DHCP Service

```
R9#show ip dhcp binding
```

Bindings from all pools not associated with VRF:			
IP address	Client-ID/ Hardware address/ User name	Lease expiration	Type
192.168.6.21	0102.4275.7961.00	Jun 09 2026 10:51 AM	Automatic
192.168.7.21	0102.4293.7ed9.00	Jun 09 2026 10:51 AM	Automatic

```
R9#
```

10.CONCLUSIONS

- Multi-area enterprise network successfully designed and implemented.
- Dynamic routing, VPN, redundancy, and network services were successfully integrated.
- All project objectives were successfully validated.

Future Improvements:

- IPv6 integration
- DMVPN for scalable VPN deployments
- Network automation using Ansible
- Advanced AAA services (TACACS+/RADIUS)

THANK YOU FOR YOUR ATTENTION!